Pedro Felipe de Oliveira

Software Engineer • Java & TypeScript • Distributed Systems • REST APIs • PostgreSQL
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Professional Summary

Fifth-semester Computer Science undergraduate at the University of Fortaleza (UNIFOR), expected to graduate in June 2028. Backend-focused developer with hands-on experience across two ecosystems: seven months of professional Java/Spring Boot development (REST APIs, PostgreSQL, JPA/Hibernate) at Eletra Energy Solutions, and an independently built distributed financial transaction system in TypeScript/NestJS — covering persistent idempotency, concurrency control, and event-driven architecture, validated by 297 automated tests. I also built Resíduo, an open-source full-stack tool that reproduces and provides evidence of deleted records reappearing in offline and event-driven systems. Comfortable moving between statically-typed, framework-driven back-ends (Java/Spring) and the Node.js/TypeScript ecosystem, with PostgreSQL as a common thread. Seeking an internship or entry-level backend engineering role; open to on-site or hybrid opportunities in Fortaleza and remote roles in Brazil or internationally.

Education

University of Fortaleza — UNIFOR — Fortaleza, CE, Brazil
Bachelor of Science in Computer Science — *Feb 2024 – Jun 2028 (expected)*

- Fifth semester, morning classes, with an overall GPA of 7.11/10.

Experience

Eletra Energy Solutions — Fortaleza, CE, Brazil (Hybrid)
Software Development Fellow — *Feb 2024 – Aug 2024 (7 months)*

- Developed and delivered a client-server application for hierarchical lookup of energy meter product lines, categories, and models.
- Built a REST API with CRUD operations using Java 8 and Spring Boot, with PostgreSQL persistence through JPA/Hibernate and Swagger documentation.
- Built a JavaFX/FXML interface, integrated the client with the API, and created and maintained unit tests with JUnit and Mockito for the core application.
- Contributed to a Maven and Git/GitFlow workflow, including feature branches, pull requests, code reviews, continuous integration (CI/CD), and Agile ceremonies.

Projects

Distributed Wagering Processor — ***Jungle Gaming Backend Challenge*** — NestJS, Bun, TypeScript, PostgreSQL, AWS SQS, Docker

- Designed and independently built a distributed financial service that processes betting transactions (BET, WIN, LOSS, REFUND, ROLLBACK) via HTTP and a FIFO SQS queue, guaranteeing a never-negative balance and an immutable accounting ledger.
- Implemented persistent idempotency across two layers (Idempotency-Key + Inbox pattern) and real concurrency control via pessimistic locking (`SELECT ... FOR UPDATE`), validated with tests spawning multiple real OS processes competing for the same wallet.
- Applied the Outbox pattern to publish events only after the financial commit, with exponential backoff and a worker that automatically retries out-of-order references.
- Modeled the domain with DDD, decoupled from framework and ORM, replicating business invariants (non-negative balance, uniqueness, ledger immutability) directly as PostgreSQL constraints and triggers.
- Covered the system with 297 test cases (unit, integration, and concurrency with real processes) across roughly 14k lines of TypeScript, plus Prometheus metrics, structured logs, and liveness/readiness health checks.

Resíduo — ***Personal Project*** — ***Open Source*** TypeScript, NestJS, React, PostgreSQL, Playwright

- Built, end to end, a lab that reproduces against a real service the bug of deleted records reappearing through delayed offline updates and message redeliveries (*upsert*), with verdicts based solely on reads taken after a confirmed deletion.
- Implemented the fixed version with the Inbox pattern, per-identity pessimistic locking, and per-generation tombstones in PostgreSQL, plus deterministic orchestration through explicit barriers (no *sleeps*) and reproducible JSON/Markdown reports.
- Built the React interface (event timeline, per-step HTTP evidence, side-by-side mode comparison) and validated the system with integration tests against real PostgreSQL, a real concurrency test, and Playwright E2E tests in CI (GitHub Actions).

My City, My Street — ***Academic Project*** React Native, Expo, Firebase

- Team-built a mobile app for reporting urban incidents, with Firebase Authentication, real-time listings via Cloud Firestore, and image upload via Firebase Storage.
- Implemented an offline queue with AsyncStorage and automatic sync on reconnect, Expo Router navigation, and an interactive city map.

Skills

- **Java & Spring:** Java, Spring Boot, JPA/Hibernate, Maven, JavaFX, Swagger
- **TypeScript & Node:** TypeScript, NestJS, Node.js, Bun, MikroORM, React
- **Distributed Systems:** Idempotency, Concurrency (Pessimistic Locking), Event-Driven Architecture, Outbox/Inbox Pattern, AWS SQS, Domain-Driven Design (DDD), Offline Sync, Observability (Prometheus, Health Checks)
- **Cross-Stack & Tools:** REST APIs, Object-Oriented Programming, PostgreSQL, SQL, Firebase, Git/GitFlow, JUnit, Mockito, Unit Testing, Integration and E2E Testing (Vitest, Playwright), CI/CD (GitHub Actions), Docker
- **Languages:** Portuguese (Native), English (Pre-Intermediate / A2), Spanish (Basic)

Additional Information

- **Interests:** back-end development, REST APIs, web systems, distributed systems, and software quality.
- **Availability:** open to full-time or part-time opportunities, including remote roles in Brazil or internationally.